

Traditional faces 100% Avoid waters — check conditions before departure.

Warning 100.0% | Caution 0.0% | Wind max 18.6 kt



Traditional — current conditions

Suitable Caution Avoid



Traditional

100% Avoid

AVOID

0% Caution



Very Small Motor

100% Avoid

AVOID

0% Caution



Small Craft

0% Avoid

CAUTION

100% Caution



Larger Vessels

0% Avoid

SUITABLE

0% Caution

Wind up to 19 kt · Seas up to 1.6 m

Primary driver: None

Fleet: AVOID
most sensitive class

High risk for small traditional vessels; larger vessels retain mostly suitable windows.



SAME OCEAN - DIFFERENT RISK

Suitability varies significantly by vessel class under identical sea conditions

Suitable Caution Avoid

Same forecast moment: Mon, 06 Jul 2026 21:00:00 UTC — selected for maximum vessel-class contrast.

Traditional

0% Suitable



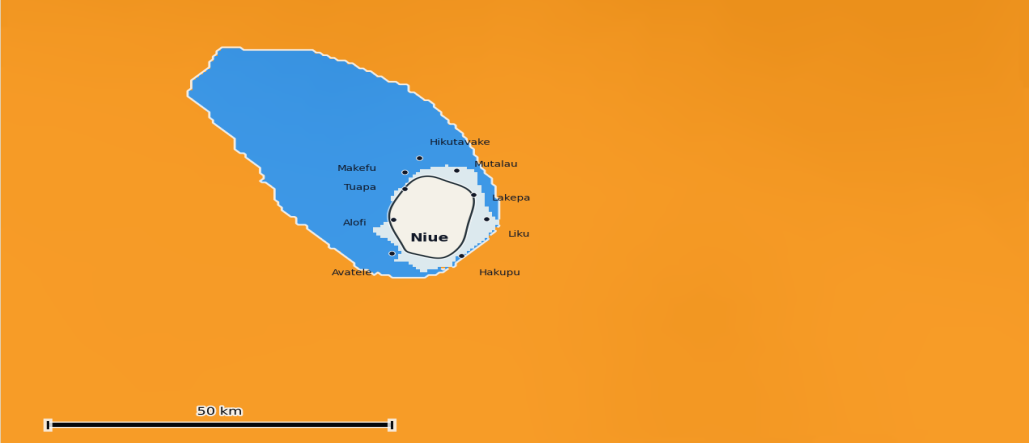
0% Suitable

0% Caution

0% Avoid

Very Small Motor

0% Suitable



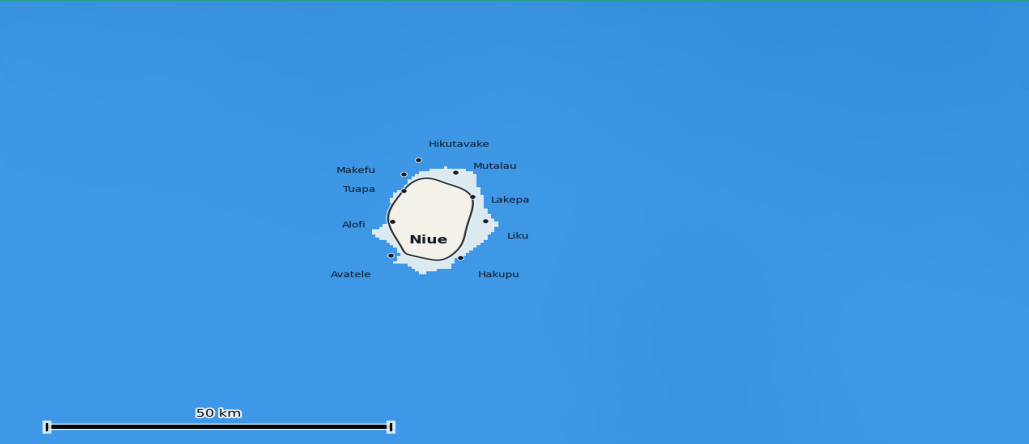
0% Suitable

0% Caution

0% Avoid

Small Craft

0% Suitable



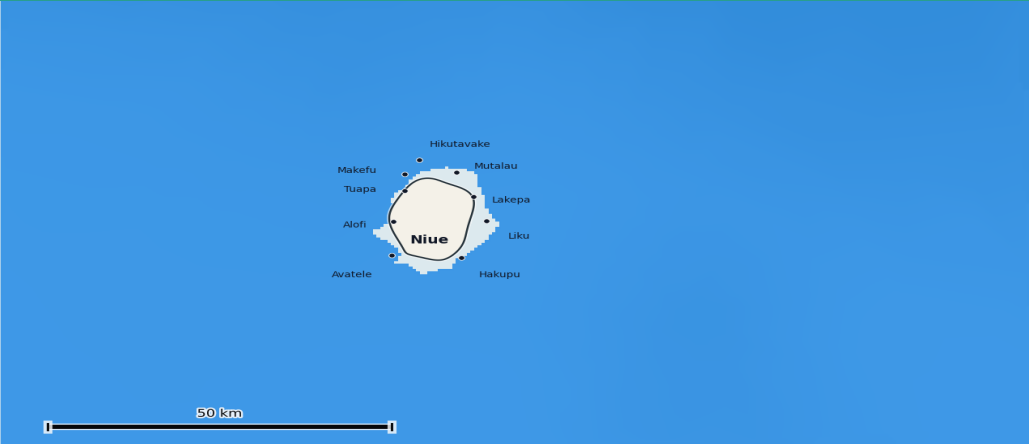
0% Suitable

0% Caution

0% Avoid

Larger Vessels

0% Suitable



0% Suitable

0% Caution

0% Avoid

Dashed rectangle on each map = SWAN model domain boundary. No suitability classification is available outside this extent.

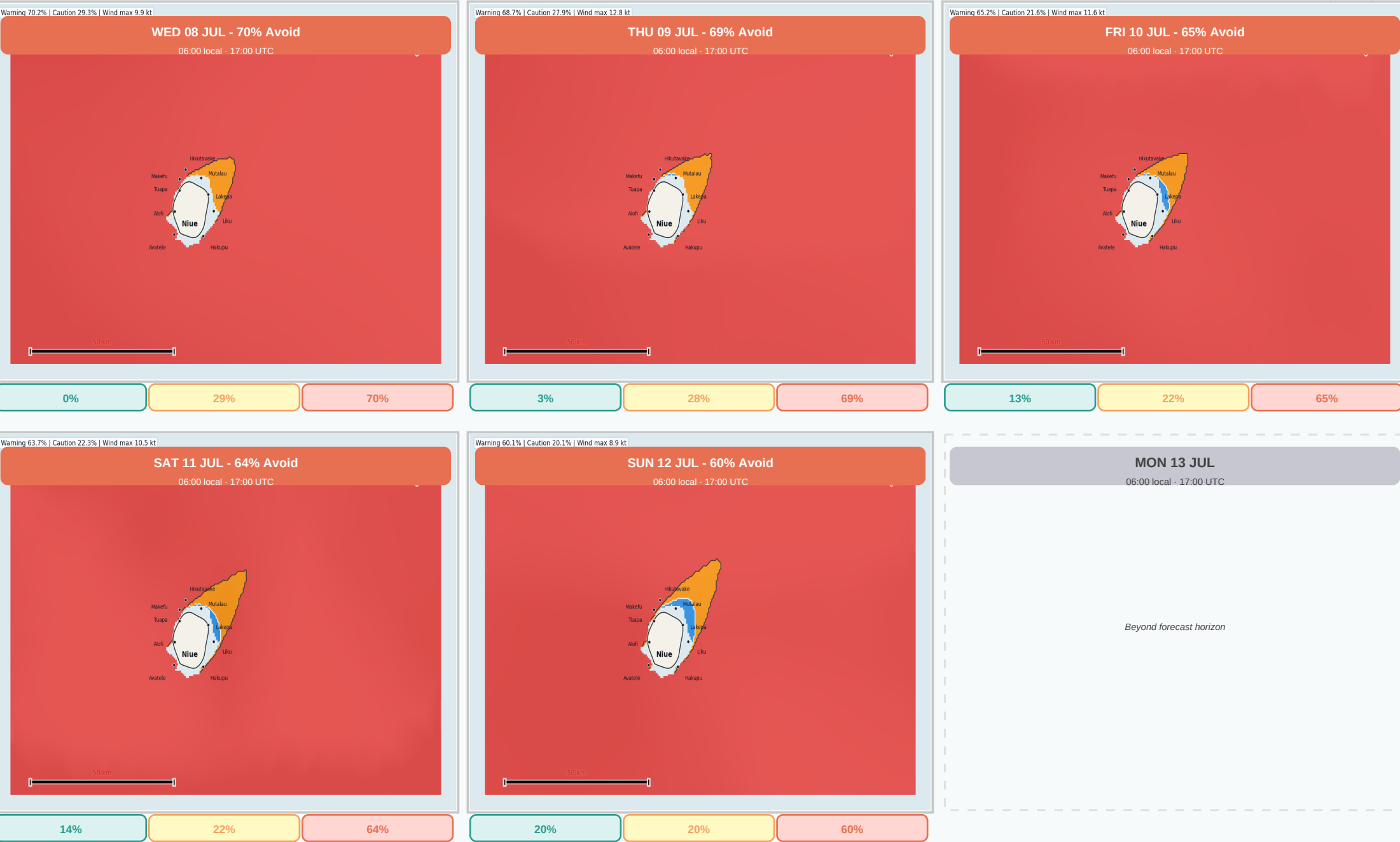
SWAN wave model guidance — mesh resolution may not capture all local sea state variability. Use alongside official warnings.

TRADITIONAL - FORECAST EVOLUTION

Daily 06:00 Niue local time (17:00 UTC) — 5-day evolution

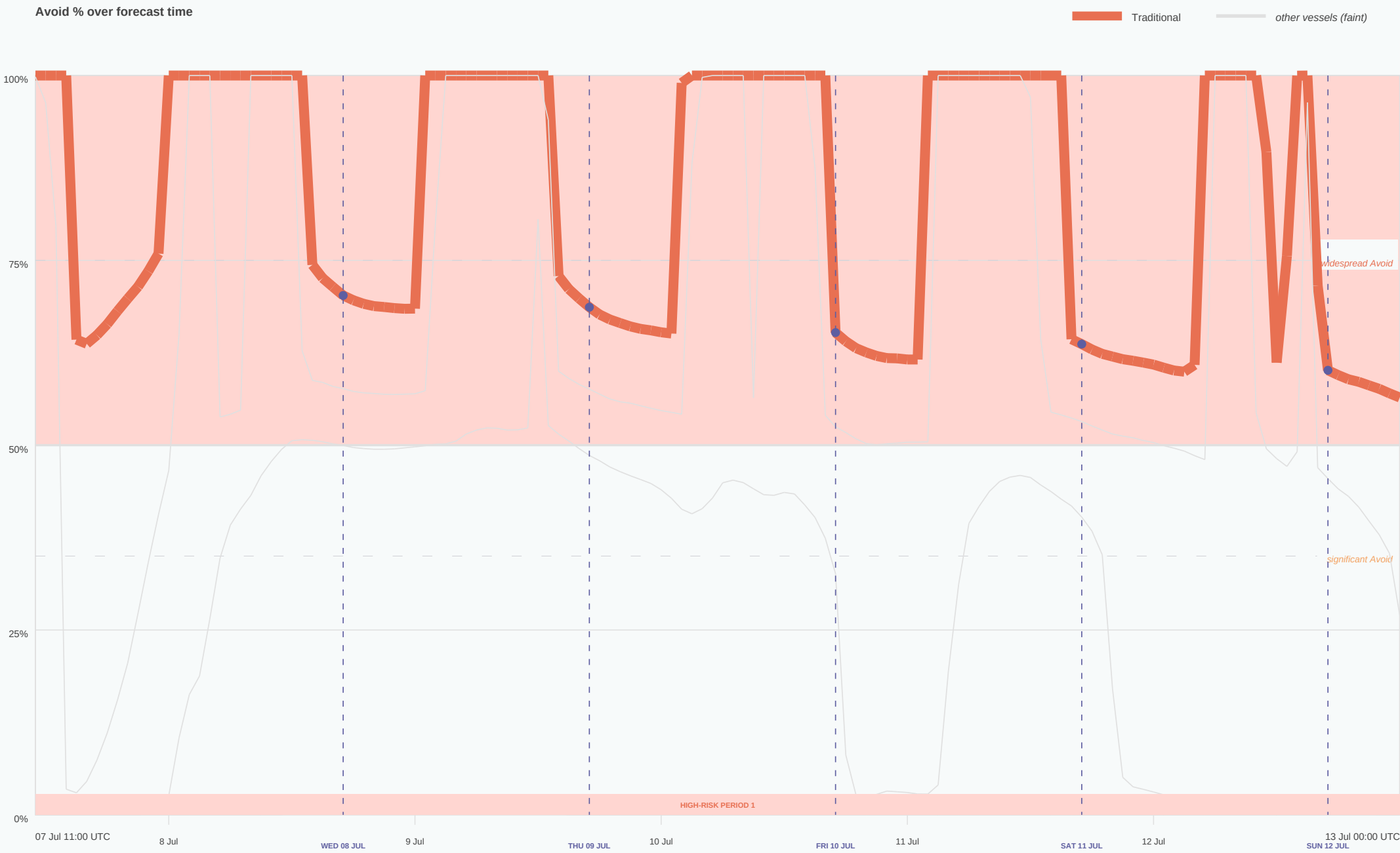
Traditional conditions briefly improve, then deteriorate sharply to full Avoid before recovering later.

Bottom bars: Suitable | Caution | Avoid



TRADITIONAL - FORECAST TREND

Avoid % across the full forecast period — all vessel classes



Traditional conditions briefly improve, then deteriorate sharply to full Avoid before recovering later.

"Recovery window" = period between consecutive high-risk events where Avoid% drops below 50% of the domain.
SWAN wave model guidance — mesh resolution may not capture all local sea state variability. Use alongside official warnings.

HOW SUITABILITY IS CLASSIFIED

Classification is applied independently to each grid cell using two meteorological parameters: significant wave height (Hs) and sustained 10-metre wind speed. A cell is flagged Avoid when either parameter meets or exceeds the warning threshold for the selected vessel class; Caution when it meets the caution threshold; Suitable otherwise. Domain-level summaries report the percentage of grid cells in each category. Thresholds are defined per vessel class as shown in the table below.

Vessel Class	Typical Use	Caution Threshold	Avoid Threshold
Traditional Craft	Canoes, vaka, outrigger canoes	Hs >= 0.5 m Wind >= 10 kt	Hs >= 1.0 m Wind >= 12 kt
Very Small Motor	Dinghies, open skiffs under 6 m	Hs >= 1.0 m Wind >= 12 kt	Hs >= 1.5 m Wind >= 15 kt
Small Craft	Fibreglass boats 6-10 m, inter-island skiffs	Hs >= 1.5 m Wind >= 15 kt	Hs >= 2.0 m Wind >= 20 kt
Larger Vessels	Decked vessels over 10-12 m	Hs >= 2.5 m Wind >= 20 kt	Hs >= 3.0 m Wind >= 25 kt

Classification:

Suitable — all parameters below caution thresholds

Caution — any parameter at or above caution threshold

Avoid — any parameter at or above warning threshold

WAVE MODEL

SWAN (Simulating WAVes Nearshore) — third-generation spectral wave model run operationally by the Pacific Community (SPC) for Niue. Atmospheric forcing: GFS 0.25° analysis fields.

FORECAST SCHEDULE

Runs every 6 hours: 00:00, 06:00, 12:00, 18:00 UTC. Forecast horizon: 7 days.

SEA-LEVEL CONTEXT

Page 2 shows total sea level derived from astronomical tide, inverse barometer, and sea-level anomaly. This is contextual water-level guidance only and is not currently used in the wind/wave vessel suitability classification.

LIMITATIONS & CAVEATS

- Outputs are from a numerical model; local conditions may differ.
- Forecast uncertainty grows at longer lead times (>48 h).
- Coastal effects (currents, swell refraction) may not be fully resolved.
- Not a substitute for official maritime warnings — always cross-check with Niue Met Service.

ATTRIBUTION

Produced by the Pacific Community (SPC), Ocean & Maritime Programme.
SWAN © Delft University of Technology. Thresholds per WMO Pacific guidelines.
Advisory scope — Vessel: Traditional · Window: 7-day outlook · Area: Full model domain